

pH Experimental Plan and **Results**

Titration Experiment Summary

- 6 repeats
- 5 Solutions with a nitrogen supply
- Nitrogen won't affect pH and is used until the scrubbers are ready
- Measure pH after titration and volume consumption of titrant
- Assess against the technical requirements [1, 2, 3, 4]

CO2 PPM Experiment Summary

- 7 Repeats using a CO2 scrubber
- Scrubbers are filled each day with a filter either side, 20ml capacity
- Constant flow rate of 1l/min
- CO2 sensor will monitor the output
- Sensors have an accuracy of +/-50ppm
- pH starts to be affected if 50ppm > is maintained for 5 or more minutes.
- Assess against the technical requirements [5]

Future Work

- Upon successful completion and meeting of the technical requirements of the PPM and titration project, the two experiments will be combined to confirm no unwanted interactions occur when delivered in the final product.

Appendix A – Technical Requirements

- 1) Average pH of each solution = 8 ± 0.1
- 2) Standard deviation of the pH of each solution ≤ 0.1
- 3) Maximum difference in average pH between solutions ≤ 0.1
- 4) Volume consumption of NaOH during titration $\leq 5\text{ml}$ per solution in every run
- 5) Solution pH is maintained for 6 hours*

**Critical that this is maintained in all test runs due to the assay being most sensitive to pH*

Appendix B – Titration Experiment Data

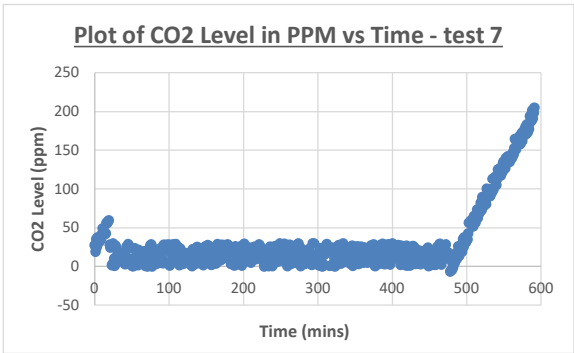
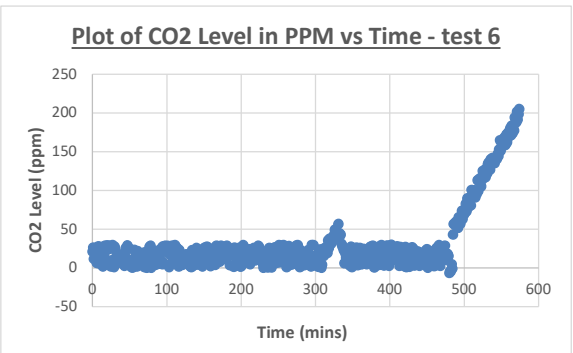
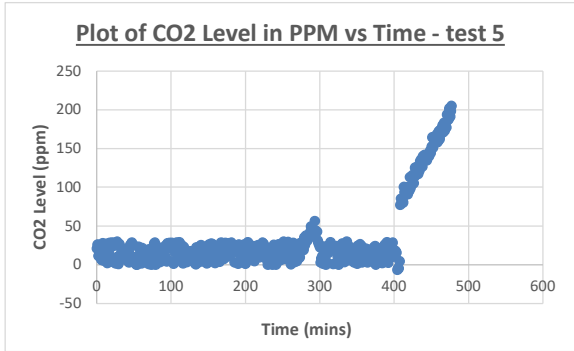
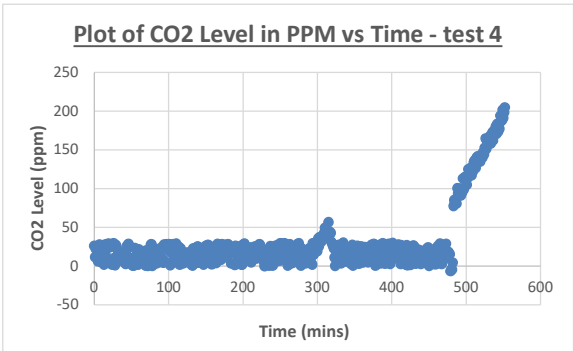
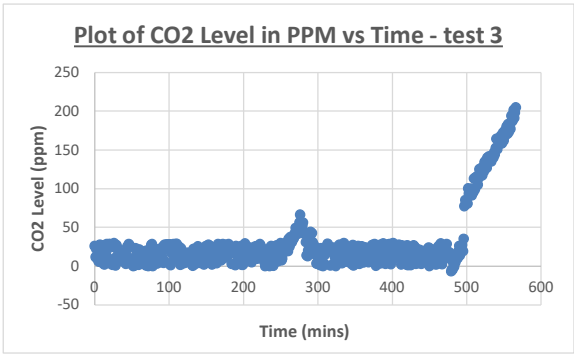
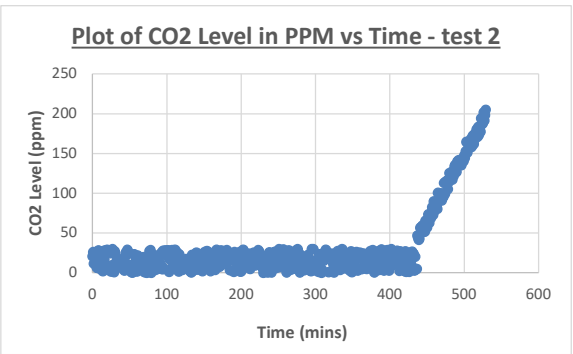
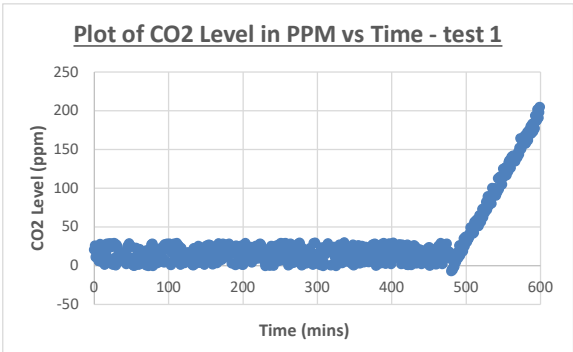
Table 1: Titration pH data summary (n=6) with average and standard deviation data

n	Solution A	Solution B	Solution C	Solution D	Solution E
1	7.90	8.00	7.95	8.00	7.95
2	7.85	8.05	7.95	8.00	7.95
3	8.00	8.10	8.20	7.80	8.05
4	8.10	7.90	8.00	8.20	8.05
5	7.95	8.00	8.05	8.00	8.00
6	8.00	8.00	7.90	8.00	7.90
Average	7.97	8.01	8.01	8.00	7.98
STD	0.08	0.06	0.10	0.12	0.06

Table 2: Titration sodium hydroxide consumption data summary (n=6) with average and standard deviation data

n	Solution A	Solution B	Solution C	Solution D	Solution E
1	4.0	3.0	4.5	4.6	4.6
2	4.5	3.3	4.5	4.8	4.5
3	5.0	3.0	4.6	4.6	4.3
4	4.6	3.5	4.4	4.7	4.5
5	4.7	4.0	4.0	5.0	4.6
6	4.5	3.6	4.5	4.5	4.7
Average	4.55	3.40	4.42	4.70	4.53
STD	0.30	0.35	0.20	0.16	0.12

Appendix C - CO2 raw data plots



239	6.57	6.57	6.57	6.57	6.57	6.57	6.57
240	4.56	4.56	4.56	4.56	4.56	4.56	4.56
241	0.47	0.47	0.47	0.47	0.47	0.47	0.47
242	15.14	15.14	15.14	15.14	15.14	15.14	15.14
243	13.5	13.5	13.5	13.5	13.5	13.5	13.5
244	25.89	25.89	25.89	25.89	25.89	25.89	25.89
245	16.08	16.08	16.08	16.08	16.08	16.08	16.08
246	26.63	26.63	26.63	26.63	26.63	26.63	26.63
247	6.11	6.11	6.11	6.11	6.11	6.11	6.11
248	17.59	17.59	17.59	17.59	17.59	17.59	17.59
249	16.23	16.23	16.23	16.23	16.23	16.23	16.23
250	8.36	8.36	8.36	8.36	8.36	8.36	8.36
251	29.58	29.58	29.58	29.58	29.58	29.58	29.58
252	7.37	7.37	7.37	7.37	7.37	7.37	7.37
253	10.03	10.03	10.03	10.03	10.03	10.03	10.03
254	28.56	28.56	28.56	28.56	28.56	28.56	28.56
255	26.03	26.03	26.03	26.03	26.03	26.03	26.03
256	22.69	22.69	22.69	22.69	22.69	22.69	22.69
257	29.51	29.51	29.51	29.51	29.51	29.51	29.51
258	17.69	17.69	18.97	17.69	17.69	17.69	17.69
259	25.51	25.51	22.35	25.51	25.51	25.51	25.51
260	13.17	13.17	27.29	13.17	13.17	13.17	13.17
261	0.63	0.63	19.25	0.63	0.63	0.63	0.63
262	27.35	27.35	35.42	27.35	27.35	27.35	27.35
263	5.45	5.45	25.35	5.45	5.45	5.45	5.45
264	28.21	28.21	37.56	28.21	28.21	28.21	28.21
265	5.63	5.63	33.67	5.63	5.63	5.63	5.63
266	3.47	3.47	31.01	3.47	3.47	3.47	3.47
267	17.94	17.94	33.6	17.94	17.94	17.94	17.94
268	14.96	14.96	35.06	14.96	14.96	14.96	14.96
269	4.67	4.67	36.95	4.67	4.67	4.67	4.67
270	22.17	22.17	42.43	22.17	22.17	22.17	22.17
271	21.69	21.69	49.28	21.69	21.69	21.69	21.69
272	8.91	8.91	46.95	8.91	8.91	8.91	8.91
273	9.78	9.78	44.57	9.78	9.78	9.78	9.78
274	28.22	28.22	41.59	28.22	28.22	28.22	28.22
275	4.71	4.71	42.84	4.71	18.97	4.71	4.71
276	29.64	29.64	56.53	29.64	22.35	29.64	29.64
277	25.1	25.1	66.53	25.1	27.29	25.1	25.1
278	24.53	24.53	56.53	24.53	19.25	24.53	24.53
279	0.63	0.63	50.53	0.63	35.42	0.63	0.63
280	22.28	22.28	52.53	22.28	25.35	22.28	22.28
281	9.56	9.56	56.53	9.56	37.56	9.56	9.56
282	14.98	14.98	44.57	14.98	33.67	14.98	14.98
283	15.61	15.61	41.59	15.61	31.01	15.61	15.61
284	14.55	14.55	42.84	14.55	33.6	14.55	14.55
285	3.32	3.32	31.01	3.32	35.06	3.32	3.32
286	13.93	13.93	13.93	13.93	36.95	13.93	13.93

335	7.91	7.91	7.91	7.91	7.91	31.01	7.91
336	30	30	30	30	30	28.56	30
337	9.7	9.7	9.7	9.7	9.7	26.03	9.7
338	24.99	24.99	24.99	24.99	24.99	22.69	24.99
339	5.8	5.8	5.8	5.8	5.8	5.8	5.8
340	18.52	18.52	18.52	18.52	18.52	18.52	18.52
341	13.27	13.27	13.27	13.27	13.27	13.27	13.27
342	16.75	16.75	16.75	16.75	16.75	16.75	16.75
343	13.55	13.55	13.55	13.55	13.55	13.55	13.55
344	5.73	5.73	5.73	5.73	5.73	5.73	5.73
345	20.71	20.71	20.71	20.71	20.71	20.71	20.71
346	3.05	3.05	3.05	3.05	3.05	3.05	3.05
347	15.43	15.43	15.43	15.43	17.66	15.43	15.43
348	27.23	27.23	27.23	27.23	19.22	27.23	27.23
349	0.78	0.78	0.78	0.78	6.57	0.78	0.78
350	11.42	11.42	11.42	11.42	1.25	11.42	11.42
351	18.67	18.67	18.67	18.67	24.54	18.67	18.67
352	6.54	6.54	6.54	6.54	16.03	6.54	6.54
353	6.29	6.29	6.29	6.29	9.88	6.29	6.29
354	16.78	16.78	16.78	16.78	26.68	16.78	16.78
355	26.22	26.22	26.22	26.22	24.52	26.22	26.22
356	19.67	19.67	19.67	19.67	24.24	19.67	19.67
357	3.14	3.14	3.14	3.14	3.1	3.14	3.14
358	5.06	5.06	5.06	5.06	20.33	5.06	5.06
359	20.93	20.93	20.93	20.93	4.27	20.93	20.93
360	25.62	25.62	25.62	25.62	4.92	25.62	25.62
361	16.97	16.97	16.97	16.97	15.31	16.97	16.97
362	4.82	4.82	4.82	4.82	20.91	4.82	4.82
363	4.97	4.97	4.97	4.97	18.84	4.97	4.97
364	23.87	23.87	23.87	23.87	2.78	23.87	23.87
365	24.26	24.26	24.26	24.26	10.95	24.26	24.26
366	22.84	22.84	22.84	22.84	12.48	22.84	22.84
367	14.38	14.38	14.38	14.38	14.19	14.38	14.38
368	12.92	12.92	12.92	12.92	16.4	12.92	12.92
369	4.82	4.82	4.82	4.82	5.98	4.82	4.82
370	17.54	17.54	17.54	17.54	11.37	17.54	17.54
371	24.06	24.06	24.06	24.06	20.18	24.06	24.06
372	17.04	17.04	17.04	17.04	21.98	17.04	17.04
373	18.82	18.82	18.82	18.82	21.95	18.82	18.82
374	27.4	27.4	27.4	27.4	24.17	27.4	27.4
375	13.17	13.17	13.17	13.17	0.24	13.17	13.17
376	16.06	16.06	16.06	16.06	11.81	16.06	16.06
377	28.65	28.65	28.65	28.65	16.81	28.65	28.65
378	4.99	4.99	4.99	4.99	12.8	4.99	4.99
379	0.42	0.42	0.42	0.42	20.44	0.42	0.42
380	9.82	9.82	9.82	9.82	19.86	9.82	9.82
381	18.94	18.94	18.94	18.94	2.73	18.94	18.94
382	15.93	15.93	15.93	15.93	17.68	15.93	15.93

383	3.27	3.27	3.27	3.27	21.83	3.27	3.27
384	7.27	7.27	7.27	7.27	2.09	7.27	7.27
385	27.51	27.51	27.51	27.51	3.41	27.51	27.51
386	3.69	3.69	3.69	3.69	14.92	3.69	3.69
387	23.27	23.27	23.27	23.27	25.69	23.27	23.27
388	17.02	17.02	17.02	17.02	22.77	17.02	17.02
389	29.41	29.41	29.41	29.41	21.67	29.41	29.41
390	12.62	12.62	12.62	12.62	29.25	12.62	12.62
391	10.41	10.41	10.41	10.41	23.78	10.41	10.41
392	3	3	3	3	2.55	3	3
393	25.83	25.83	25.83	25.83	10.74	25.83	25.83
394	24.13	24.13	24.13	24.13	17.89	24.13	24.13
395	19.2	19.2	19.2	19.2	10.02	19.2	19.2
396	28.29	28.29	28.29	28.29	15.34	28.29	28.29
397	28.6	28.6	28.6	28.6	20.13	28.6	28.6
398	24.97	24.97	24.97	24.97	28.63	24.97	24.97
399	2.66	2.66	2.66	2.66	10.27	2.66	2.66
400	28	28	28	28	15.13	28	28
401	10.41	10.41	10.41	10.41	18.3	10.41	10.41
402	29.81	29.81	29.81	29.81	17.23	29.81	29.81
403	9.41	9.41	9.41	9.41	15.35	9.41	9.41
404	14.24	14.24	14.24	14.24	-6.63	14.24	14.24
405	15.29	15.29	15.29	15.29	-4.44	15.29	15.29
406	19.88	19.88	19.88	19.88	-4.66	19.88	19.88
407	26.4	26.4	26.4	26.4	4.58	26.4	26.4
408	15.39	15.39	15.39	15.39	77.4	15.39	15.39
409	20.76	20.76	20.76	20.76	85.21	20.76	20.76
410	27.82	27.82	27.82	27.82	80.51	27.82	27.82
411	25.7	25.7	25.7	25.7	83.63	25.7	25.7
412	19.91	19.91	19.91	19.91	80.56	19.91	19.91
413	2.36	2.36	2.36	2.36	100.44	2.36	2.36
414	27.04	27.04	27.04	27.04	93.66	27.04	27.04
415	10.18	10.18	10.18	10.18	91.1	10.18	10.18
416	19.75	19.75	19.75	19.75	91.82	19.75	19.75
417	5.35	5.35	5.35	5.35	100.06	5.35	5.35
418	15.68	15.68	15.68	15.68	91.11	15.68	15.68
419	4.47	4.47	4.47	4.47	99.5	4.47	4.47
420	1.81	1.81	1.81	1.81	95.41	1.81	1.81
421	27.07	27.07	27.07	27.07	113.09	27.07	27.07
422	1.13	1.13	1.13	1.13	99.11	1.13	1.13
423	17.66	17.66	17.66	17.66	114.49	17.66	17.66
424	19.22	19.22	19.22	19.22	113.68	19.22	19.22
425	6.57	6.57	6.57	6.57	115.94	6.57	6.57
426	1.25	1.25	1.25	1.25	105.14	1.25	1.25
427	24.54	24.54	24.54	24.54	113.55	24.54	24.54
428	16.03	16.03	16.03	16.03	125.14	16.03	16.03
429	9.88	9.88	9.88	9.88	119.82	9.88	9.88
430	26.68	26.68	26.68	26.68	120.47	26.68	26.68

431	24.52	24.52	24.52	24.52	126.78	24.52	24.52
432	24.24	24.24	24.24	24.24	117.13	24.24	24.24
433	3.1	3.1	3.1	3.1	123.77	3.1	3.1
434	20.33	20.33	20.33	20.33	121.88	20.33	20.33
435	4.27	4.27	4.27	4.27	133.91	4.27	4.27
436	4.92	4.92	4.92	4.92	135.8	4.92	4.92
437	15.31	46.95	15.31	15.31	130.07	15.31	15.31
438	20.91	44.57	20.91	20.91	126.8	20.91	20.91
439	18.84	41.59	18.84	18.84	140.25	18.84	18.84
440	2.78	42.84	2.78	2.78	134.81	2.78	2.78
441	10.95	56.53	10.95	10.95	141.98	10.95	10.95
442	12.48	55.93	12.48	12.48	136.23	12.48	12.48
443	14.19	57.8	14.19	14.19	137.85	14.19	14.19
444	16.4	59.54	16.4	16.4	135.05	16.4	16.4
445	5.98	57.85	5.98	5.98	141.36	5.98	5.98
446	11.37	51.65	11.37	11.37	140.85	11.37	11.37
447	20.18	51.83	20.18	20.18	140.2	20.18	20.18
448	21.98	65.43	21.98	21.98	147.06	21.98	21.98
449	21.95	64.72	21.95	21.95	144.21	21.95	21.95
450	24.17	56.91	24.17	24.17	152.58	24.17	24.17
451	0.24	62.16	0.24	0.24	164.59	0.24	0.24
452	11.81	73.45	11.81	11.81	150.89	11.81	11.81
453	16.81	62.84	16.81	16.81	161.95	16.81	16.81
454	12.8	70.67	12.8	12.8	163.88	12.8	12.8
455	20.44	68.85	20.44	20.44	160.93	20.44	20.44
456	19.86	71.47	19.86	19.86	160.81	19.86	19.86
457	2.73	82.4	2.73	2.73	168.2	2.73	2.73
458	17.68	72.64	17.68	17.68	158.1	17.68	17.68
459	21.83	89.64	21.83	21.83	160.06	21.83	21.83
460	2.09	77.4	2.09	2.09	172.36	2.09	2.09
461	3.41	85.21	3.41	3.41	162.17	3.41	3.41
462	14.92	80.51	14.92	14.92	174.38	14.92	14.92
463	25.69	83.63	25.69	25.69	169.09	25.69	25.69
464	22.77	80.56	22.77	22.77	176.5	22.77	22.77
465	21.67	100.44	21.67	21.67	180.36	21.67	21.67
466	29.25	93.66	29.25	29.25	171.05	29.25	29.25
467	23.78	91.1	23.78	23.78	183.48	23.78	23.78
468	2.55	91.82	2.55	2.55	173.5	2.55	2.55
469	10.74	100.06	10.74	10.74	179.53	10.74	10.74
470	17.89	91.11	17.89	17.89	177.34	17.89	17.89
471	10.02	99.5	10.02	10.02	194.07	10.02	10.02
472	15.34	95.41	15.34	15.34	186.93	15.34	15.34
473	20.13	113.09	20.13	20.13	188.01	20.13	20.13
474	28.63	99.11	28.63	28.63	201.55	28.63	28.63
475	10.27	114.49	10.27	10.27	191.17	10.27	10.27
476	15.13	113.68	15.13	15.13	198.19	15.13	15.13
477	18.3	115.94	18.3	18.3	204.97	18.3	18.3
478	17.23	105.14	17.23	17.23		17.23	17.23

479	15.35	113.55	15.35	15.35	15.35	15.35
480	-6.63	125.14	-6.63	-6.63	-6.63	-6.63
481	-4.44	119.82	-4.44	-4.44	-4.44	-4.44
482	-4.66	120.47	-4.66	-4.66	-4.66	-4.66
483	4.58	126.78	4.58	4.58	4.58	4.58
484	-1.01	117.13	-1.01	77.4	-1.01	-1.01
485	9.47	123.77	9.47	85.21	42.84	9.47
486	4.93	121.88	4.93	80.51	56.53	4.93
487	13.34	133.91	13.34	83.63	55.93	13.34
488	10.16	135.8	10.16	80.56	57.8	10.16
489	10.14	130.07	10.14	100.44	59.54	10.14
490	19.25	126.8	19.25	93.66	57.85	19.25
491	26.3	140.25	26.3	91.1	51.65	26.3
492	13.91	134.81	13.91	91.82	51.83	13.91
493	18.97	141.98	18.97	100.06	65.43	18.97
494	22.35	136.23	22.35	91.11	64.72	22.35
495	27.29	137.85	27.29	99.5	56.91	27.29
496	19.25	135.05	19.25	95.41	62.16	19.25
497	35.42	141.36	35.42	113.09	73.45	35.42
498	25.35	140.85	77.4	99.11	62.84	25.35
499	37.56	140.2	85.21	114.49	70.67	37.56
500	33.67	147.06	80.51	113.68	68.85	33.67
501	31.01	144.21	83.63	115.94	71.47	31.01
502	33.6	152.58	80.56	105.14	82.4	33.6
503	35.06	164.59	100.44	113.55	72.64	35.06
504	36.95	150.89	93.66	125.14	89.64	42.84
505	42.43	161.95	91.1	119.82	77.4	56.53
506	49.28	163.88	91.82	120.47	85.21	55.93
507	46.95	160.93	100.06	126.78	80.51	57.8
508	44.57	160.81	91.11	117.13	83.63	59.54
509	41.59	168.2	99.5	123.77	80.56	57.85
510	42.84	158.1	95.41	121.88	100.44	51.65
511	56.53	160.06	113.09	133.91	93.66	51.83
512	55.93	172.36	99.11	135.8	91.1	65.43
513	57.8	162.17	114.49	130.07	91.82	64.72
514	59.54	174.38	113.68	126.8	100.06	56.91
515	57.85	169.09	115.94	140.25	91.11	62.16
516	51.65	176.5	105.14	134.81	99.5	73.45
517	51.83	180.36	113.55	141.98	95.41	62.84
518	65.43	171.05	125.14	136.23	113.09	70.67
519	64.72	183.48	119.82	137.85	99.11	68.85
520	56.91	173.5	120.47	135.05	114.49	71.47
521	62.16	179.53	126.78	141.36	113.68	82.4
522	73.45	177.34	117.13	140.85	115.94	72.64
523	62.84	194.07	123.77	140.2	105.14	89.64
524	70.67	186.93	121.88	147.06	113.55	77.4
525	68.85	188.01	133.91	144.21	125.14	85.21
526	71.47	201.55	135.8	152.58	119.82	80.51

527	82.4	191.17	130.07	164.59	120.47	83.63
528	72.64	198.19	126.8	150.89	126.78	80.56
529	89.64	204.97	140.25	161.95	117.13	100.44
530	77.4		134.81	163.88	123.77	93.66
531	85.21		141.98	160.93	121.88	91.1
532	80.51		136.23	160.81	133.91	91.82
533	83.63		137.85	168.2	135.8	100.06
534	80.56		135.05	158.1	130.07	91.11
535	100.44		141.36	160.06	126.8	99.5
536	93.66		140.85	172.36	140.25	95.41
537	91.1		140.2	162.17	134.81	113.09
538	91.82		147.06	174.38	141.98	99.11
539	100.06		144.21	169.09	136.23	114.49
540	91.11		152.58	176.5	137.85	113.68
541	99.5		164.59	180.36	135.05	115.94
542	95.41		150.89	171.05	141.36	105.14
543	113.09		161.95	183.48	140.85	113.55
544	99.11		163.88	173.5	140.2	125.14
545	114.49		160.93	179.53	147.06	119.82
546	113.68		160.81	177.34	144.21	120.47
547	115.94		168.2	194.07	152.58	126.78
548	105.14		158.1	186.93	164.59	117.13
549	113.55		160.06	188.01	150.89	123.77
550	125.14		172.36	201.55	161.95	121.88
551	119.82		162.17	191.17	163.88	133.91
552	120.47		174.38	198.19	160.93	135.8
553	126.78		169.09	204.97	160.81	130.07
554	117.13		176.5		168.2	126.8
555	123.77		180.36		158.1	140.25
556	121.88		171.05		160.06	134.81
557	133.91		183.48		172.36	141.98
558	135.8		173.5		162.17	136.23
559	130.07		179.53		174.38	137.85
560	126.8		177.34		169.09	135.05
561	140.25		194.07		176.5	141.36
562	134.81		186.93		180.36	140.85
563	141.98		188.01		171.05	140.2
564	136.23		201.55		183.48	147.06
565	137.85		191.17		173.5	144.21
566	135.05		198.19		179.53	152.58
567	141.36		204.97		177.34	164.59
568	140.85				194.07	150.89
569	140.2				186.93	161.95
570	147.06				188.01	163.88
571	144.21				201.55	160.93
572	152.58				191.17	160.81
573	164.59				198.19	168.2
574	150.89				204.97	158.1

575	161.95	160.06
576	163.88	172.36
577	160.93	162.17
578	160.81	174.38
579	168.2	169.09
580	158.1	176.5
581	160.06	180.36
582	172.36	171.05
583	162.17	183.48
584	174.38	173.5
585	169.09	179.53
586	176.5	177.34
587	180.36	194.07
588	171.05	186.93
589	183.48	188.01
590	173.5	201.55
591	179.53	191.17
592	177.34	198.19
593	194.07	204.97
594	186.93	
595	188.01	
596	201.55	
597	191.17	
598	198.19	
599	204.97	